

1. Administrative & Study Identification

Full Study Title: A Phase Ib, Multicenter Study of VOB560 in Combination With MIK665 in Patients With Relapsed/Refractory Non-Hodgkin Lymphoma, Relapsed/Refractory Acute Myeloid Leukemia, or Relapsed/Refractory Multiple Myeloma **Short Title/Acronym:** VOB560-MIK665 Combination First in Human Trial **Protocol Number:** CVOB560A12201 (Novartis Internal ID) **NCT Number:** NCT04702425 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** VOB560 (BCL2 inhibitor) and MIK665 (MCL1 inhibitor) **Phase of Development:** Phase 1b **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the safety and tolerability of VOB560 in combination with MIK665, and to identify the maximum tolerated dose (MTD) and/or recommended doses and regimens for all indications. **Secondary Objectives:** To assess preliminary antitumor activity and pharmacokinetics (PK) of the drug combination.

2.2 Background & Rationale Patients with relapsed and refractory (R/R) acute myeloid leukemia (AML), non-Hodgkin lymphoma (NHL), and multiple myeloma (MM) require effective treatment options due to their high mortality rates. The inhibition of 2 anti-apoptotic proteins, B-cell lymphoma 2 (BCL2) and myeloid cell leukemia 1 (MCL1), has shown promise as a therapeutic approach. Two novel BH3 mimetics, VOB560 and MIK665, selectively inhibit BCL2 and MCL1, respectively. In combination, VOB560 and MIK665 demonstrated strong synergy and promoted cell death in various hematological cancer models by overcoming apoptosis resistance.

Note: The study halted recruitment in April 2023 for strategic business reasons and not due to any safety concerns.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation/expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 37 patients enrolled in the dose-escalation phase before early termination (29 AML, 4 MM, and 4 NHL). **Number of Sites:** Multicenter **Estimated**

Study Start: Early 2021 **Estimated Primary Completion:** April 2023
(Terminated)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients with relapsed and/or refractory Non-Hodgkin Lymphoma (NHL), Acute Myeloid Leukemia (AML), or Multiple Myeloma (MM). 2. For AML: Pathologically confirmed diagnosis as defined by WHO Classification with $\geq 5\%$ blasts in bone marrow who have failed all conventional therapy. 3. For MM: Treated with at least 2 prior regimens, including an IMiD, a proteasome inhibitor, and an anti-CD38 antibody.

4.2 Exclusion Criteria 1. Autologous stem cell transplant within 3 months before the first dose of study treatment or prior allogeneic stem cell transplant. 2. History of or current interstitial lung disease or pneumonitis grade ≥ 2 . 3. Impaired hepatic and renal function (e.g., AST/ALT $> 1.5 \times$ ULN, Bilirubin $> 1.5 \times$ ULN, Creatinine clearance < 50 mL/min).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: VOB560 administered weekly (QW) intravenously (IV) at 25 mg and 50 mg in combination with MIK665 administered IV at 25 mg and 50 mg QW during the dose-escalation phase. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Until disease progression, unacceptable toxicity, physician decision, or study termination.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for hematological malignancies and Tumor Lysis Syndrome (TLS) prophylaxis. **Prohibited:** Concurrent administration of other investigational anti-cancer therapies or standard re-induction chemotherapy during the study period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, Baseline Labs, Disease Assessment Baseline
Day 0	First Dose (VOB560 + MIK665 IV), PK blood sampling		
Interim	Weekly (QW)	Safety Labs, DLT Evaluation, Efficacy Assessment, Tumor Lysis Syndrome (TLS) monitoring	End of Study
Treatment End	Final Vital Signs, Exit Interview, Disease Progression confirmation, Study Termination procedures		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs) during the first treatment cycle to determine the Maximum Tolerated Dose (MTD) and/or Recommended Doses for expansion. **Measurement Method:** Clinical observation, safety laboratory parameters, and investigator assessment according to standard toxicity criteria.

7.2 Secondary & Exploratory Endpoints 1. Preliminary antitumor activity (e.g., Overall Response Rate). 2. Pharmacokinetic (PK) profiles of VOB560 and MIK665 (e.g., SC_{max} , AUC).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of Treatment-Related AEs (TRAEs). Frequent TRAEs observed included diarrhea, nausea, vomiting, pyrexia, and Tumor Lysis Syndrome (TLS). **Serious Adverse Events (SAEs):** Evaluated continuously throughout the trial with standard 24-hour reporting protocols. Severe grade ≥ 3 events observed included TLS, neutropenia, myopericarditis, and differentiation syndrome. **Data Monitoring Committee (DMC):** Internal Safety Review Committee evaluating dose-escalation decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set included all patients who received at least one dose of the investigational combination (N=37). **Sample Size Justification:** Guided by standard Phase 1 dose-escalation principles to identify MTD; actual enrollment was 37 before strategic termination. **Handling of Missing Data:** Descriptive statistics primarily utilized due to the early termination and dose-escalation nature of the Phase 1b trial.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits by Novartis or designated CRO. **Audit Trail:** Standardized electronic Case Report Form (eCRF) utilized for data capture and querying.

11. Study Identifiers & Governance

Primary ID: NCT04702425 **Registry Number:** NCT04702425 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** VOB560-MIK665 Combination First in Human Trial **Official Regulatory Title:** A

Phase Ib, Multicenter Study of VOB560 in Combination With MIK665 in Patients With Relapsed/Refractory Non-Hodgkin Lymphoma, Relapsed/Refractory Acute Myeloid Leukemia, or Relapsed/Refractory Multiple Myeloma

12. Clinical Framework (Oncology Specific)

Primary Condition: Relapsed/Refractory Hematological Malignancies (AML, NHL, MM) **Diagnosis Threshold:** Relapsed/refractory status failing prior conventional therapies (e.g., $\geq 5\%$ blasts for AML, ≥ 2 prior regimens for MM) **Medical Methodology:** Selective inhibition of BCL2 and MCL1 anti-apoptotic proteins to induce cell death **Optimization Target:** Maximum Tolerated Dose (MTD) discovery

13. Study Procedures & Methodology

Management Pathway: Dose escalation cohort design guided by DLT evaluations **Education Protocol (HCP):** Site initiation visits and protocols focusing on IV administration and TLS management/prophylaxis **Education Protocol (Patient):** Informed consent outlining Phase 1 risks, TLS symptoms, and onsite visit requirements **Quality Audit Mechanism:** Clinical monitoring and sponsor oversight

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until disease progression, death, or trial closure (Study halted in April 2023) **Onsite Visit Cadence:** Weekly corresponding to QW IV infusions **Remote Monitoring Frequency:** As necessary between treatment cycles **Checklist Review Interval:** Continuous cycle-by-cycle assessment for DLTs and TRAEs

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Investigator Name]				_____		[DD-MMM-YYYY]						
	Clinical Operations Lead				[Name]				_____		[DD-MMM-YYYY]						
	Lead Statistician				[Name]				_____		[DD-MMM-YYYY]						